

Water entry of small hydrophobic spheres on an axisymmetric wavefield

Supplemental Information

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I. DETERMINING THE OPERATING FREQUENCIES

Our heaving wavemaker is operated at frequencies between successive resonant frequencies f_r to maintain linear waves across the widest possible range of driving amplitudes A_d . When driven at or near $f = f_r$ the bath develops nonlinear waves and secondary phenomena, such as drop ejection, at lower A_d [1]. In the linear regime, theoretical models and experimental observations show excellent agreement for the interface shape, particularly at the axis of symmetry ($r = 0$) when using time-periodic solutions [1, 2].

We estimate f_r for our system as follows. First, we seek the cylindrical wave mode profile governed by

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + k^2 \right) S(r, \theta) = 0, \quad (\text{S1})$$

where the wavenumber k satisfies the deep-water dispersion relation,

$$\omega^2 = kg + \frac{\sigma k^3}{\rho}. \quad (\text{S2})$$

Here, ω is angular frequency, g is acceleration due to gravity, σ is surface tension, and ρ is fluid density. For axisymmetric modes, the surface elevation $h(r, t)$ takes the form

$$h(r, t) = a(t)S(r) = a(t)J_0(kr), \quad (\text{S3})$$

where the time-dependent amplitude $a(t)$ satisfies the harmonic oscillator equation $\frac{d^2 a}{dt^2} + \omega^2 a = 0$, yielding solutions of the form $a(t) = A_0 \sin(\omega t + \varphi)$ in steady state. Applying the boundary condition $h(R_r, t) = A_d \cos(\omega t)$ at the boundary corresponding to the inner radius R_r of the drive ring driven at an amplitude A_d , we find $A_0 = A_d / J_0(kR_r)$ and $\varphi = 0$. Consequently, the steady-state surface profile within the circular domain is estimated to be

$$h(r, t) = \frac{A_d}{J_0(kR_r)} J_0(kr) \sin(\omega t) = A_0 J_0(kr) \sin(\omega t). \quad (\text{S4})$$

The resonant frequencies are identified by the poles of this solution, which occur when the denominator vanishes, i.e., $J_0(kR_r) = 0$ wherein $A_0 \rightarrow \infty$. Solving this condition yields a discrete set of resonant wavenumbers k_r . Substituting these values back into the dispersion relation (Eq. S2) and using $\omega_r = 2\pi f_r$ produces the discrete spectrum of resonant frequencies f_r . To avoid these

resonances within our experimental operating window, the driving frequencies f were chosen as the midpoints between subsequent f_r , resulting in the test frequencies: $f = 6.79, 8.18, 9.64, 11.19, 12.86, 14.64$, and 16.54 Hz. The wave amplitude at the moment of impact A was directly measured in experiment rather than being assumed from any theoretical considerations (see Sec. VI). As such, the simplified theory presented in this section was used for two purposes only: (i) as a guide to identify suitable operating frequencies and (ii) to motivate the derivation of our wave-slope parameter m used for data collapse.

II. IMAGE PROCESSING

High-speed recordings of the cavity dynamics are processed using a custom MATLAB program that determines the cavity shape and volume $V(t)$ in each frame, the submarine pinchoff time t_p , and the pinchoff depth z_p . To establish the impact time ($t = 0$), the sphere is tracked using an above-water recording and defined as the time midway between the frames immediately before and after the south pole of the sphere touches the bath. In cases where the south pole of the sphere is appreciably closer to the free surface compared to the preceding and subsequent frames, that frame is used as the contact time $t = 0$. Using synchronized recordings of the cavity, the waterline is identified by the discontinuous shape formed by the sphere and its reflection in the frames immediately following contact.

To determine the cavity shape and volume, each frame is processed following the procedure shown in Figure S1(a). First, pixels above the pre-defined waterline are removed, and a reference background image is subtracted to isolate the cavity from the background (middle). The image is then binarized using a global threshold to suppress noise. Finally, the cavity edges are detected, a morphological closing operation is applied to connect adjacent pixels, and a filling operation encloses the remaining internal spaces. This workflow yields the segmented cavity shown on the black backdrop in the rightmost image of Figure S1(a). By looping this procedure through all frames, a complete history of the cavity shape is attained (Figure S1(b)). Assuming the cavity is axisymmetric, we determine the axis of rotation by finding the midpoint between the leftmost and rightmost edge of the 2D cavity shape at each pixel height. The cavity volume $V(t)$ is then reconstructed by treating the 2D cavity shape as a volume of revolution. Specifically, the profile is discretized into horizontal slices of single-pixel height, and the total volume is determined by integrating these individual slices using the disk method. The final volume includes the air and

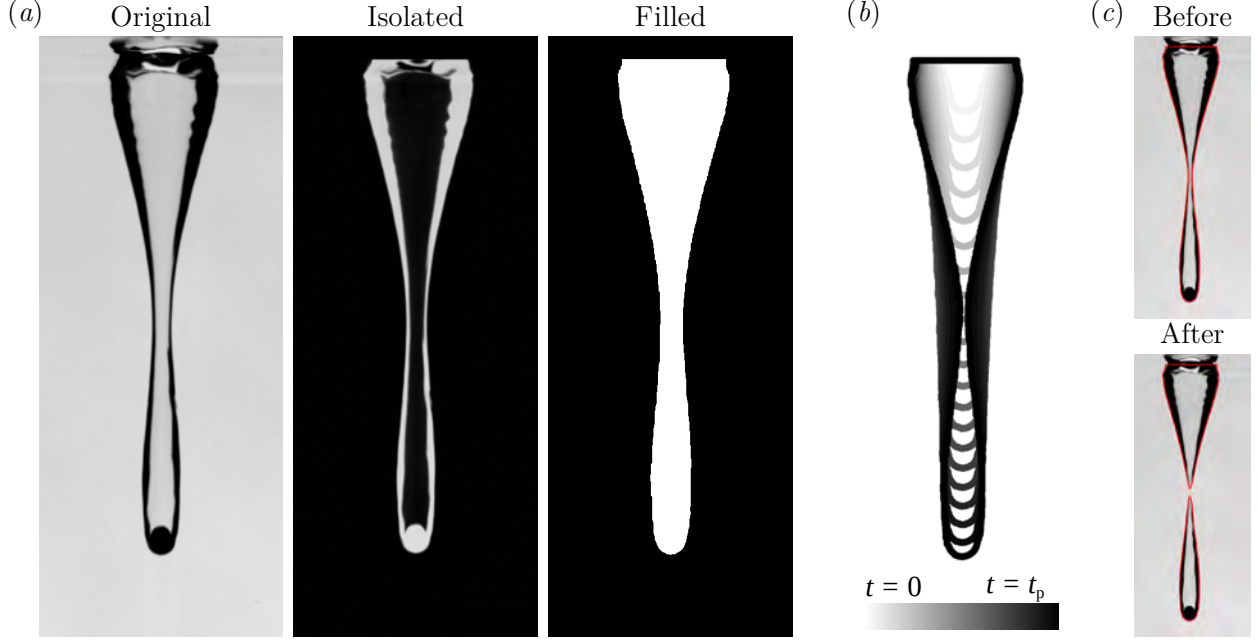


Figure S1: (a) Steps in the image processing procedure. (b) Cavity outlines detected for evenly spaced times up to pinchoff $t = t_p$. (c) Detected single and double cavity volumes in the frame before (top) and after (bottom) pinchoff.

sphere volume, although the latter is negligibly small.

The cavity shape and volume are tracked until submarine pinchoff, at which point two additional metrics are recorded. To identify the exact moment of pinchoff, the code identifies the frames before and after the air cavity splits into two segments, as shown in Figure S1(c). The pinchoff time t_p is defined as the time occurring halfway between these frames relative to the initial impact time. The pinchoff depth z_p is the distance from the waterline to the midpoint of the gap separating the two cavity segments in the frame immediately following pinchoff. For a few of our highest- We cases, the downward-moving jet produced by surface seal interacted with the cavity near the pinchoff location, requiring us to manually determine the pinchoff frame.

II.1. Uncertainty quantification

The spatial and temporal resolutions of the imaging system establish the baseline instrumental uncertainties. The above-water and submarine recordings have spatial resolutions of 0.0313 mm/pixel and 0.0588 mm/pixel, respectively, and both record at a frame rate of $f_{ps} = 7500$ frames per second (giving an inter-frame interval of 0.133 ms).

The pinchoff time t_p is determined from the duration between two discrete events: initial impact

and submarine pinchoff. Assuming the frame for each event is identified within ± 1 frame under a uniform uncertainty distribution, the standard uncertainty for individual frame selection is

$$u_{F_s} = u_{F_e} = \frac{1}{\sqrt{3}} \text{ frames.} \quad (\text{S5})$$

Propagating these independent uncertainties yields a total uncertainty for t_p of

$$u_{t_p} = \sqrt{\left(\frac{u_{F_e}}{fps}\right)^2 + \left(\frac{u_{F_s}}{fps}\right)^2} = 0.109 \text{ ms.} \quad (\text{S6})$$

This instrumental uncertainty remains slightly below the single-frame temporal resolution of 0.133 ms.

The uncertainty in the pinchoff depth z_p arises from two primary factors: spatial camera resolution and frame selection timing. Assuming a uniform distribution for a ± 1 pixel detection threshold, the spatial resolution limits depth accuracy to $u_{z_p,1} = 0.048$ mm. The uncertainty stemming from the discrete frame rate depends on the vertical velocity of the pinchoff location, modeled as

$$u_{z_p,2} = \left| \frac{dz_p}{dt} \right| u_t. \quad (\text{S7})$$

While the exact rate of change $|dz_p/dt|$ is unknown, we can estimate an upper bound using the maximum impact velocity tested $U = 5.6$ m/s, yielding $u_{z_p,2} \approx U \cdot u_{t_p}$. Combining these independent sources in quadrature gives the maximum total uncertainty for the pinchoff depth

$$u_{z_p} = \sqrt{(u_{z_p,1})^2 + (u_{z_p,2})^2} = 0.434 \text{ mm.} \quad (\text{S8})$$

The cavity volume V is reconstructed via the disk method by assuming axisymmetry and integrating over N vertical pixel slices as described previously. Potential sources of volumetric uncertainty include cross-frame volume variations, departures from axisymmetry, and edge-detection accuracy. We can safely ignore volume variation, as the change in volume $|\frac{dV}{dt}|$ slows near pinchoff for deep seals and is approximately zero at pinchoff for surface seals (cf. Figure 4). Additionally, uncertainty due to asymmetry is minimal since wavefield and impactor are axisymmetric. The primary source of uncertainty stems from measuring the cavity radius R_i , which is mathematically coupled to tracking of the centerline. The standard uniform distribution for a ± 1 pixel detection

threshold for both cavity edges across an individual slice is $u_{r_i} = 0.0244$ mm. The total propagated volume uncertainty u_V is

$$u_V = \sqrt{\sum_{i=1}^N (V u_{r_i})^2} = \sqrt{\sum_{i=1}^N (2\pi R_i u_{r_i})^2}. \quad (\text{S9})$$

To approximate an upper bound on this uncertainty, we assume negligible sphere deceleration such that $N \approx U \cdot t_p / s$ (where s is the spatial calibration factor in mm/pixel). Evaluating this bound at the maximum impact speed ($U = 5.6$ m/s) for the longest deep-seal cavities ($t = t_p$) with a uniform radius $R_i \approx R$ yields a total volume uncertainty of less than 1%.

These systematic instrumental uncertainties are negligibly small compared to the statistical variance observed between repeat trials. Consequently, the error bars reported throughout this study represent the standard deviation calculated across three identical experimental runs.

III. INFLUENCE OF A_d AND f ON THE CAVITY SHAPE AT CLOSURE

Increasing the driving amplitude A_d or frequency f enhances the influence of waves on the cavity seal type and resulting pinchoff metrics (see Figure 3 in the main text). Figure S2 illustrates this for a sphere ($We = 560$; $Bo = 0.54$) impacting a wave crest ($\varphi = \pi/2$) for (a) increasing A_d , and (b) increasing f . In both cases, the cavity expands with the control parameter and transitions from a surface seal to deep seal. Once in the deep seal regime, further increases in the control parameter have a negligible effect on the pinchoff metrics, though they noticeably increase the height and width of the splash crown. Similar visualizations showing the influence of the wave phase φ on the cavity shape are provided in Figure 1(f) of the main text. The videos accompanying these images, as well as complementary videos showing the same impacts on wave troughs are provided as Movies 5-8 in the Supplemental Material.

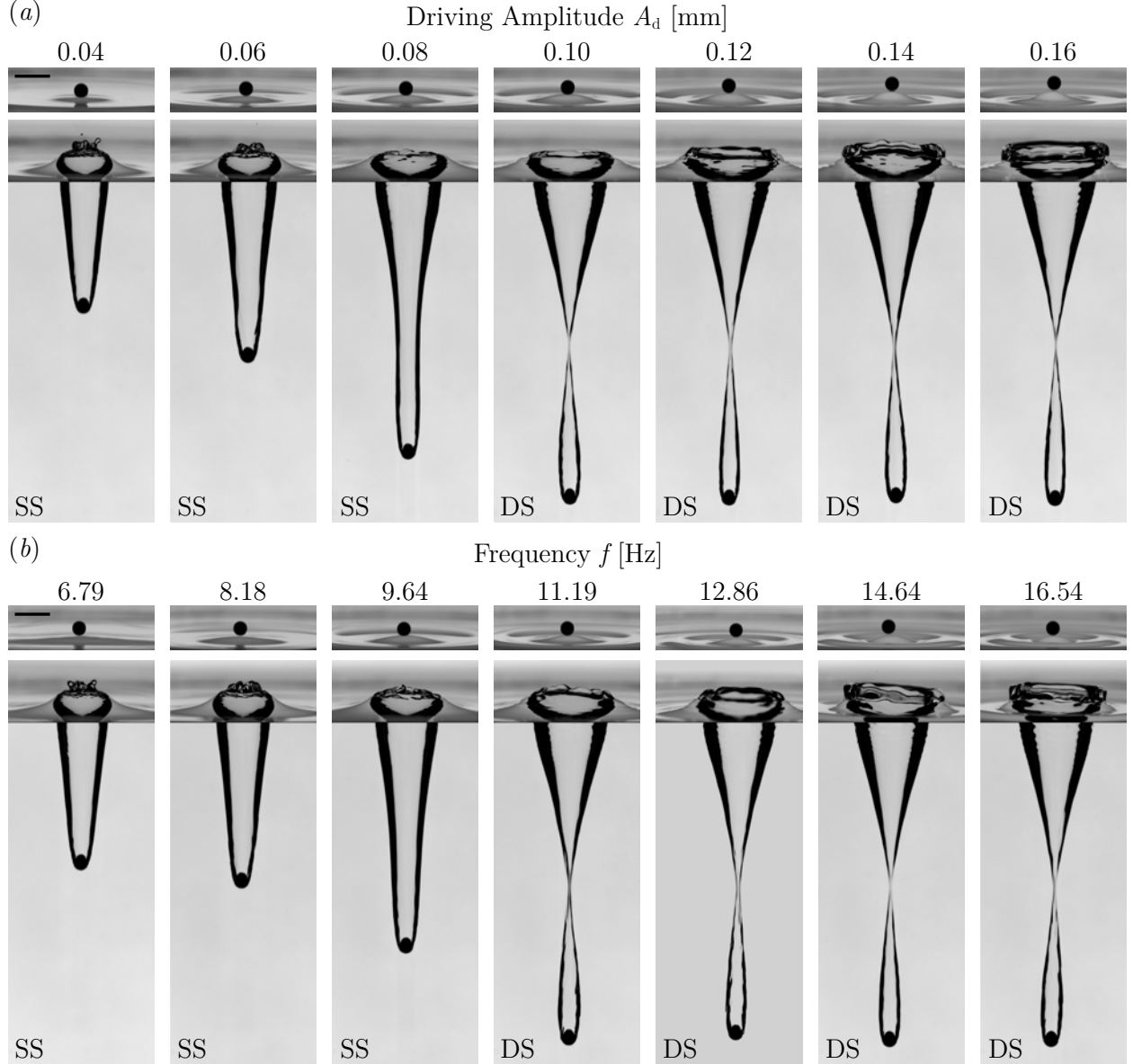


Figure S2: Sphere impact ($We = 560$; $Bo = 0.54$) on a wave crest ($\varphi = \pi/2$) at impact ($t = 0$; top) and cavity closure ($t = t_c$; bottom) for increasing (a) driving amplitude A_d and (b) frequency f . Deep seal and surface seals are annotated DS and SS, respectively. Scale bars are 10 mm.

IV. INFLUENCE OF We AND φ ON CAVITY SHAPE AT CLOSURE

Figure S3 presents experimental images corresponding to the data presented in Fig. 2 in the main text. The (a) still-water, (b) crest ($\varphi = 2/\pi$), and (c) trough ($\varphi = 3\pi/2$) cases are contrasted across seven Weber numbers We , which increase from left to right and span the deep-to-surface seal transition. The transition We and cavity shapes at lower We are the same for the still-water and trough cases, though the cavities for the trough case are slightly reduced in the surface seal

regime. For the crest case, the transition We increases. This extends the positive trend between cavity volume and We within the deep seal regime, forming larger cavities relative to the other two wave configurations (cf. $We = 490$). The videos associated with these images are provided as Movies 2-4 in the Supplemental Material.

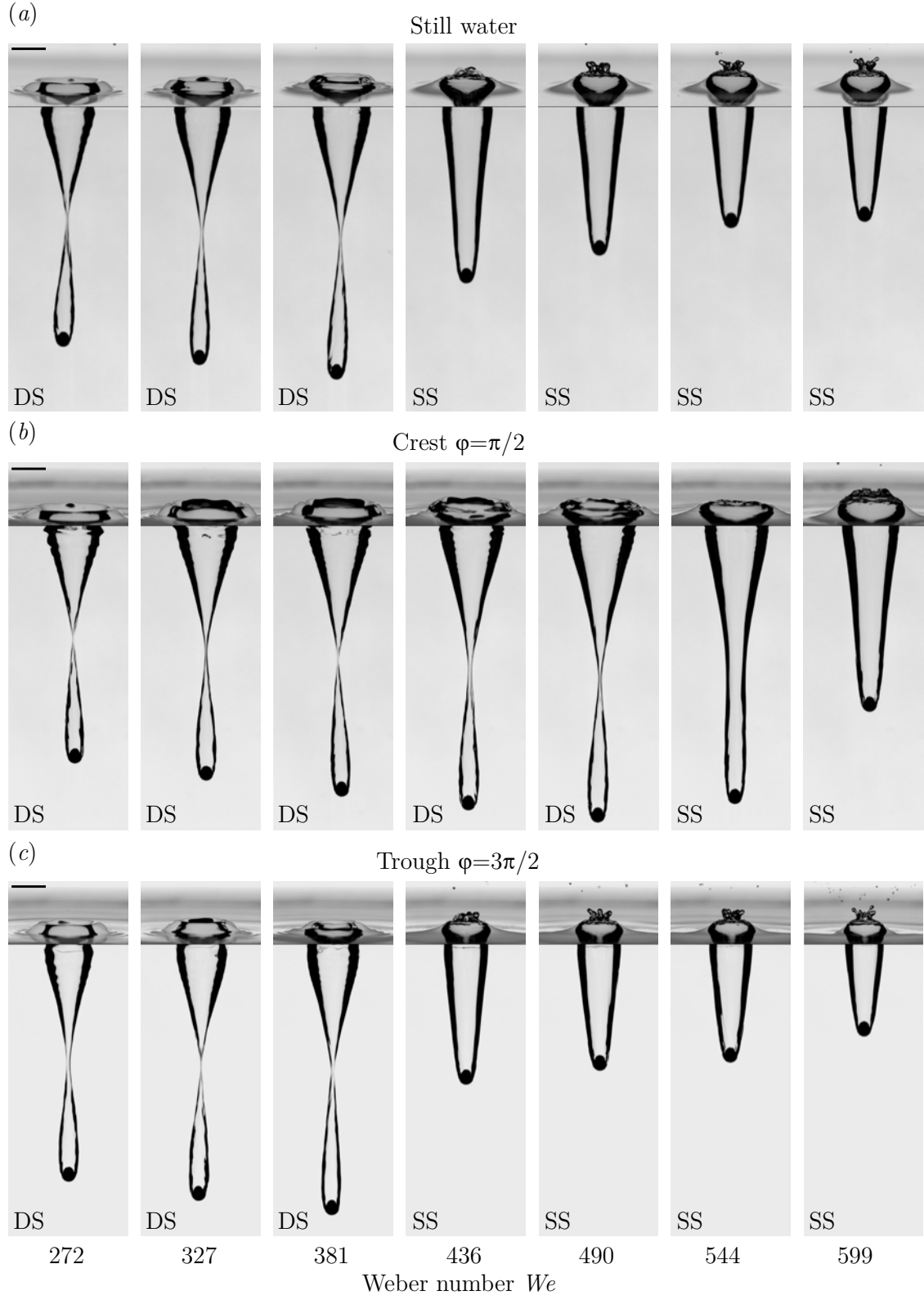


Figure S3: Cavity closure ($t = t_c$) for sphere impact ($Bo = 0.54$) on (a) still water, (b) a wave crest ($\varphi = \pi/2$), and (c) a wave trough ($\varphi = 3\pi/2$). Weber number We increases from left to right and deep seal and surface seals are annotated DS and SS, respectively. Scale bars are 10 mm long.

V. INFLUENCE OF We AND Bo ON CAVITY METRICS

V.1. Cavity metrics for $We = 560$, $Bo = 0.54$

In the main article (Figure 3), we demonstrate how our three wave-based control parameters (φ , f , A_d) influence the maximum volume of air entrained during water entry. In Figure S4, we extend this analysis to the dimensionless pinchoff depth $z_p^* = z_p/R$ ($a - c$) and pinchoff time $t_p^* (d - f)$ for $We = 560$ and $Bo = 0.54$. In each panel, deep seal and surface seal cases are differentiated by open and filled symbols, and the mean and standard deviation for impacts on still water are the dashed line and gray band, respectively. Because $We = 560$ exceeds the transition value differentiating deep and surface seal for $Bo = 0.54$, all instances featuring negative or zero wave amplitudes $A \leq 0$ close via a surface seal. For $A > 0$, waves drive a transition from deep to surface seal as f or A_d increases. The corresponding images for the data in (a, d) are included in the main text (Figure 1(f)), and those for the crest cases in (b, c, e, f) are shown in Figure S2.

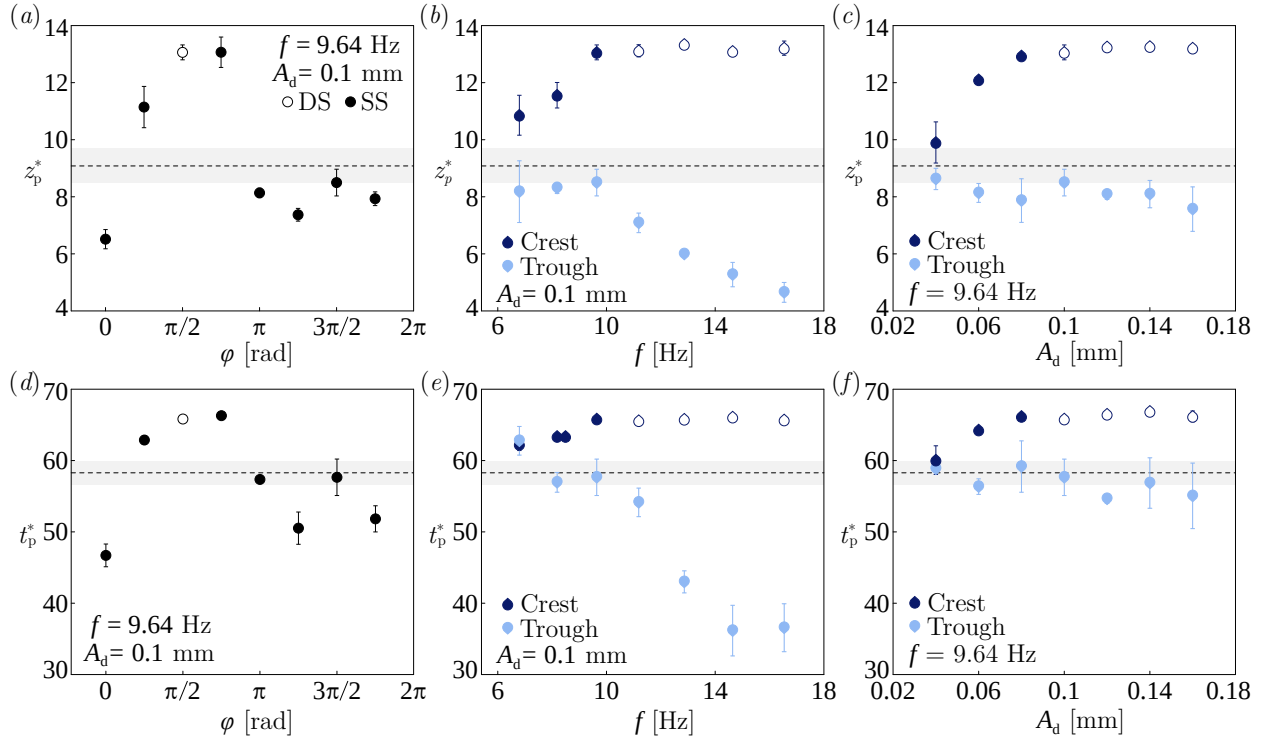


Figure S4: $(a - c)$ Dimensionless pinchoff depth $z_p^* = z_p/R$ and $(d - f)$ pinchoff time $t_p^* = t_p U/R$ as it depends upon (a, d) phase φ , (b, e) frequency f , and (c, f) driving amplitude A_d for $We = 560$ and $Bo = 0.54$. The dashed line and shaded area denote the mean and standard deviation for still-water impacts. Open and filled symbols represent deep seals (DS) and surface seals (SS), respectively.

V.2. Cavity metrics for $We = 367$, $Bo = 0.54$

When the Weber number We is lower than the threshold required for a surface seal on still water, waves with negative amplitudes can accelerate the inward collapse of the splash crown and induce a surface seal transition. Figure S5 shows the influence of phase ϕ (a, d, g), frequency f (b, e, h), and driving amplitude A_d (c, f, i) on the maximum cavity volume $V^* = V_m/V_s$ ($a - c$), pinchoff depth z_p^* ($d - f$), and pinchoff time t_p^* ($g - i$) for $We = 367$ and $Bo = 0.54$. In each panel, deep seal and surface seal cases are differentiated by open and filled symbols, and the mean and standard deviation for impacts on still water are the dashed line and gray band, respectively. In contrast to the higher- We experiments discussed in Section V.1, negative-amplitude waves here drive a transition from a deep seal to a surface seal as f or A_d increases, whereas positive-amplitude waves consistently produce deep seals. Furthermore, at this lower We the overall variation in V^* , z_p^* , and t_p^* across the parameter sweeps is noticeably reduced. This compressed range is a consequence of the increased number of deep seal closures, where the cavity metrics are less sensitive to changes in the control parameters (see Figure S2).

V.3. Cavity metrics for $We = 560$, $Bo = 1.06$

Increasing the Bond number Bo raises the critical We required to transition to surface seal. Figure S6 shows the influence of phase ϕ (a, d, g), frequency f (b, e, h), and driving amplitude A_d (c, f, i) on the maximum cavity volume $V^* = V_m/V_s$ ($a - c$), pinchoff depth z_p^* ($d - f$), and pinchoff time t_p^* ($g - i$) for $We = 560$ and $Bo = 1.06$. In each panel, deep seal and surface seal cases are differentiated by open and filled symbols, and the mean and standard deviation for impacts on still water are the dashed line and gray band, respectively. Most notably, the metrics associated with deep seals have a lower standard deviation compared to those for surface seals, an observation consistent with the trends for $We = 367$, $Bo = 0.54$, shown previously (Figure S5). Upon a detailed review of the high-speed videos, we attribute this increased variability among surface-seal trials to the downward liquid jet produced during the collapse and closure of the splash crown, which perturbs the pinchoff depth and timing in a manner that is difficult to fully control.

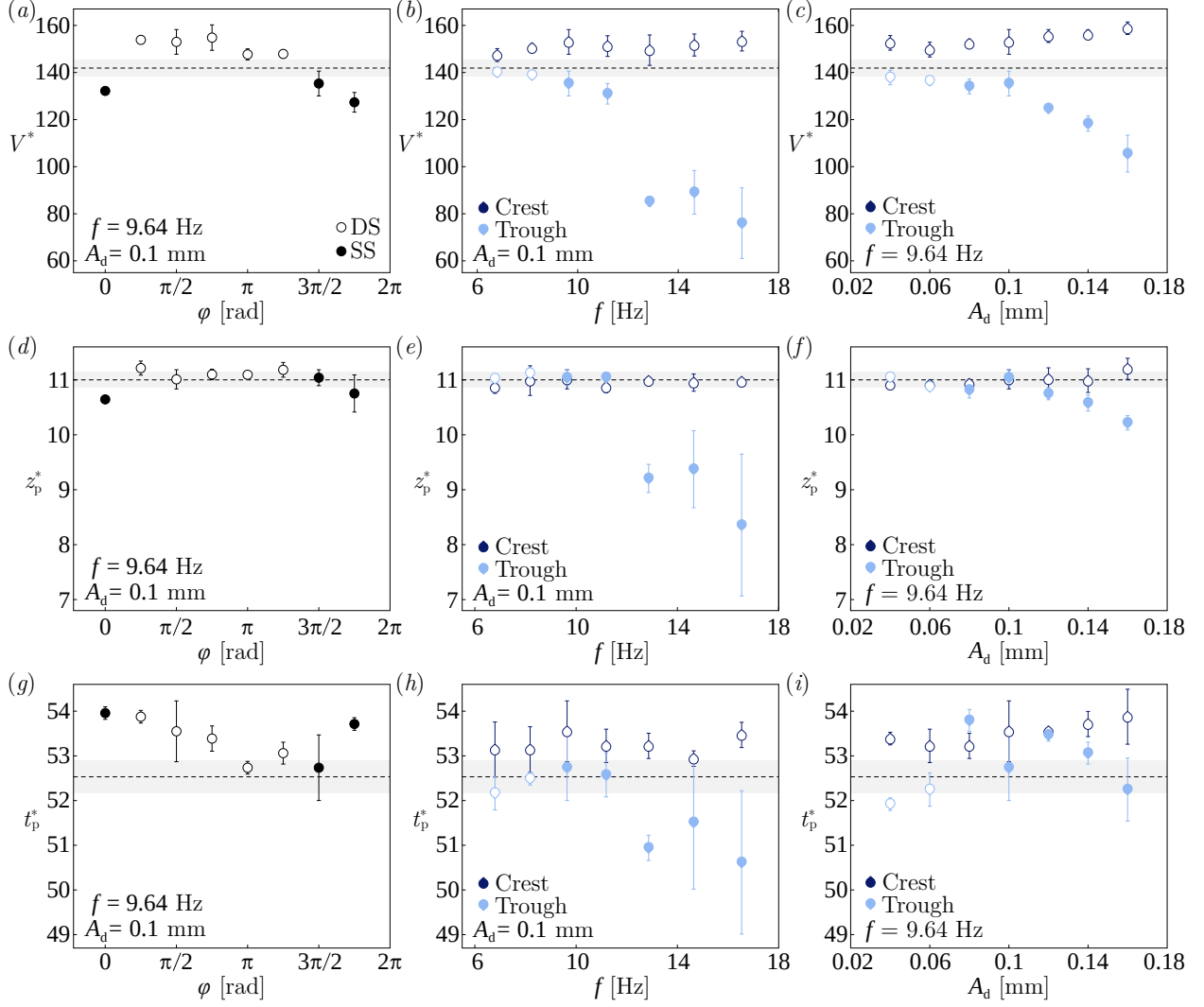


Figure S5: ((a – c) Dimensionless maximum cavity volume $V^* = V_m/V_s$, (d – f) pinchoff depth $z_p^* = z_p/R$ and (g – i) pinchoff time $t_p^* = t_p U/R$ as it depends upon (a, d, g) phase ϕ , (b, e, h) frequency f , and (c, f, i) driving amplitude A_d for $We = 367$ and $Bo = 0.54$. The dashed line and shaded area denote the mean and standard deviation for still-water impacts. Open and filled symbols represent deep seals (DS) and surface seals (SS), respectively.

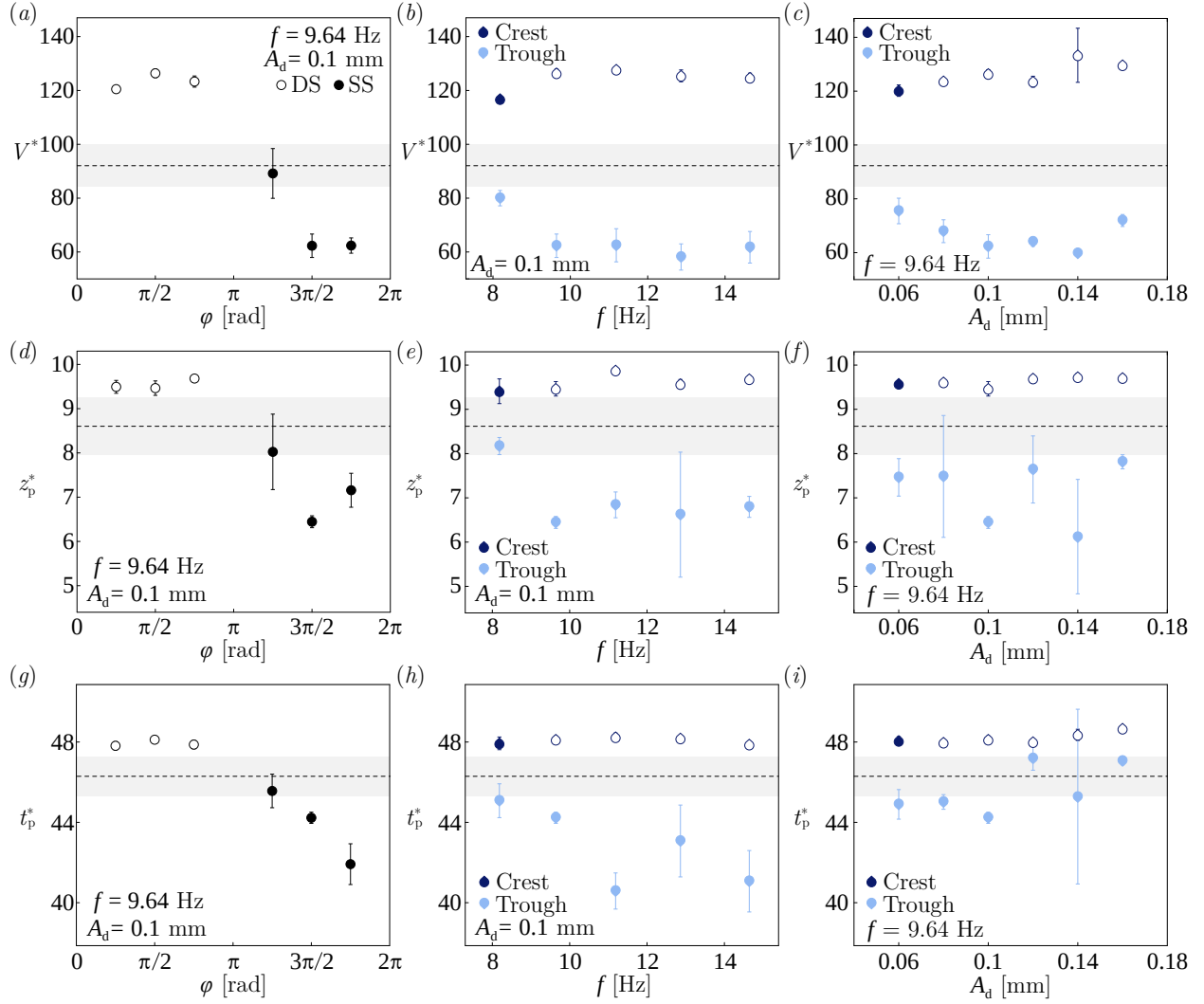


Figure S6: (a – c) Dimensionless maximum cavity volume $V^* = V_m/V_s$, (d – f) pinchoff depth $z_p^* = z_p/R$ and (g – i) pinchoff time $t_p^* = t_p U/R$ as it depends upon (a, d, g) phase ϕ , (b, e, h) frequency f , and (c, f, i) driving amplitude A_d for $We = 560$ and $Bo = 1.06$. The dashed line and shaded area denote the mean and standard deviation for still-water impacts. Open and filled symbols represent deep seals (DS) and surface seals (SS), respectively.

VI. MEASURED WAVE AMPLITUDE A

The wave amplitude at impact A for a given set of control parameters (φ, f, A_d) is defined as the difference between the height of south pole of the sphere at the moment of impact $t = 0$ and its height at impact with no waves. Figure S7 shows the measured amplitude normalized by sphere radius $A^* = A/R$ against (a) phase φ , (b) frequency f , and (c) driving amplitude A_d , where deep seal and surface seal cases are differentiated by open and filled markers, respectively. In each panel, the remaining two control parameters are held constant, and the dashed line and gray horizontal band show the average and standard deviation across the three reference still-water experiments. Figure S7(a), alongside the corresponding experimental images in the main text (Figure 1(f)), demonstrates our capacity to achieve impact wave amplitudes whose magnitudes vary in close alignment with the prescribed phase. As shown in Figure S7(b), A^* increases marginally with f . Conversely, the driving amplitude A_d has a positive, nearly linear influence on A^* , as shown in Figure S7(c). For the data presented in panels (b) and (c), the corresponding experimental images for the crest impacts ($\varphi = \pi/2$) can be referenced in the Supplemental Information (Figure S2).

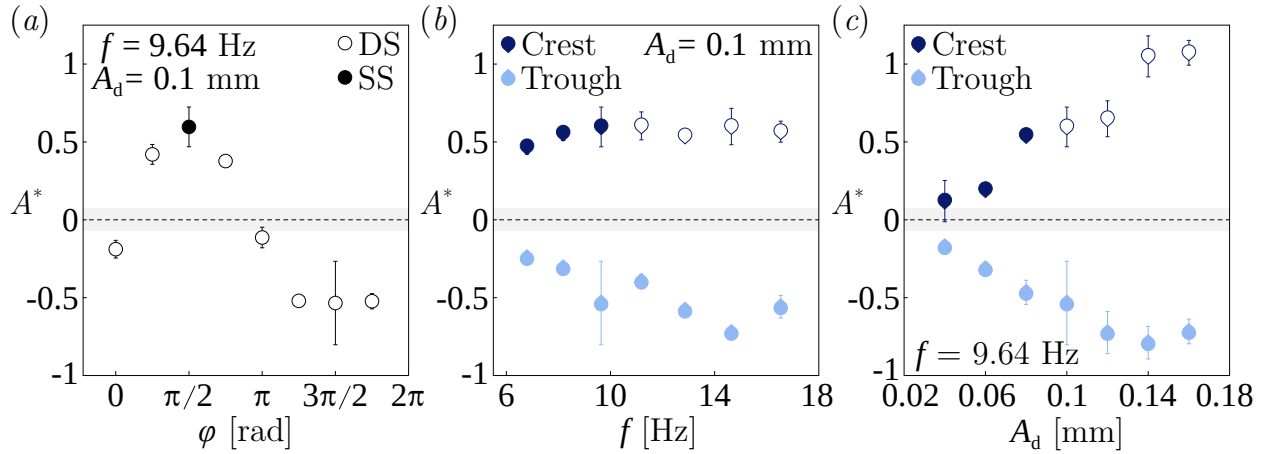


Figure S7: Dimensionless wave amplitude $A^* = A/R$ plotted against (a) phase φ , (b) frequency f , and (c) driving amplitude A_d for $We = 560$ and $Bo = 0.54$. The dashed line and shaded area denote the mean and standard deviation for still-water impacts. Open and filled symbols represent deep seals (DS) and surface seals (SS), respectively.

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